

Precision Crop Advisory System & Post-Harvest Loss Reduction Planner for Smallholder Farmers

PROBLEM ABSTRACT

India's smallholder farmers — operating plots under 2 hectares and accounting for over 85% of the farming community — receive the least personalised advisory support, while India loses an estimated 15–20% of its agricultural produce to post-harvest deterioration driven by poor storage decisions, delayed market linkage, and lack of real-time price intelligence. Participants must build an integrated AgriTech prototype that (a) ingests simple farmer-provided inputs (field location, crop type, sowing date, recent weather observation, and an optional crop/leaf photo) and returns three ranked, actionable advisories for the next 7 days covering irrigation scheduling, fertiliser application, and pest or disease alert with confidence indicators and farmer-accessible plain-language rationale, and (b) accepts crop type, quantity, current storage condition, and farmer location to recommend an optimal sell/store/transport action using mocked or real mandi price data, estimated spoilage curves, and transport cost models — with an optional SMS/WhatsApp price-threshold alert simulation.

EXPECTED OUTCOMES

- Multi-input intake form: location (pin/GPS), crop type, sowing date, weather observation, optional leaf photo
- 3 ranked advisories (irrigation / fertiliser / pest alert) for next 7 days with confidence indicators (High / Medium / Low) and plain-language rationale
- Optional: integration with open weather API (OpenWeatherMap / IMD) and leaf disease classifier using lightweight CNN
- Crop + storage + location intake with spoilage-risk classifier and Sell / Store / Transport decision engine with expected return estimate
- Mocked or real mandi price feed (Agmarknet or synthetic) with spoilage curve visualisation and transport cost model
- Optional: SMS/WhatsApp alert simulation for price threshold triggers; prototype covers at least 4 crop types and 5 location combinations

EVALUATION CRITERIA

Evaluation Criteria	Description
Understanding of the Problem Statement	Clear articulation of the problem, its context, and the pain points it addresses.
Proposed Approach	Methodology, intended tech stack, workflow, and reasoning behind design decisions.
Market Fit & Real-World Relevance	How the prototype solves a tangible real-world problem, the target audience it serves, and what makes the solution viable, scalable, and impact-driven.

IMPORTANT NOTICE — PRE-SCREENING ROUND

As part of the pre-screening round, each team is required to submit a document covering the following:

1. **Understanding of the Problem Statement:** A clear articulation of the problem, its context, and the pain points it addresses.
2. **Proposed Approach:** Your methodology, intended tech stack, workflow, and the reasoning behind your design decisions.
3. **Market Fit & Real-World Relevance:** How your prototype solves a tangible real-world problem, the target audience it serves, and what makes your solution viable, scalable, and impact-driven.

Kindly ensure that your submission is a well-structured and original **PPT Document**, reflective of your team's collective thinking.

P.S. — GitHub Submission (Highly Recommended)

It would be highly beneficial if your team additionally submits an early prototype via a public GitHub repository, with regular and visible commits throughout the pre-screening period. Consistent progress on GitHub demonstrates your team's seriousness, accelerates your ideation-to-execution cycle, and gives the evaluation panel deeper insight into your problem-solving approach.

The date and time for the online evaluation will be shared with you separately via email. Our team will review your submission and notify the application status via email.

For any queries, feel free to reach out to the organizing team.

◆ Teams selected in the prescreening round will receive their full hackathon problem statement at the TetraTHON 2026 Inauguration Ceremony — July 31, 2026 at Navrachana University. ◆